Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech
Code : 103601

Subject: Power System – II (Operation and Control)

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct answer the following (Any seven question only):*** **[2 x 7= 14]**

(a) The inertia constant of two machines are respectively 30 and 42 MJ per MVA. When the change in frequency due to change in load in first machine is 12 Hz, the change in the second machine frequency will be
(i) 10 Hz (ii) 12 Hz (iii) 15 Hz (iv) 42 Hz

(b) Maximum short circuit current occurs due to
(i) L-L-L-G fault
(ii) L-L fault
(iii) L-G fault
(iv) L-L-G fault

(c) A 11 kV, 10 MVA alternator has impedance of 0.10 pu when referred to its ratings as bases. The new value for base as 110 kV, 20 MVA will be
(i) 0.002 pu (ii) 0.20 pu (iii) 0.10 pu (iv) 0.02 pu

(d) The negative sequence component of voltage is equal to
(i) Positive sequence component with reverse direction
(ii) Same as positive sequence component
(iii) Complex conjugate of positive sequence component
(iv) None of these

(e) The per unit impedances of a transformer referred from the primary and secondary side are
(i) equal
(ii) Inversely proportional to square of transformation ratio of transformer
(iii) Proportional to square of transformation ratio of transformer
(iv) None of the above

(f) Convergence of power flow is always assured in case of
(i) Newton-Raphson method
(ii) Gauss Method
(iii) Gauss-Seidal method
(iv) Both (i) and (iii)

(g) Load flow study is carried out for
(i) Fault calculation
(ii) Stability study
(iii) System planning
(iv) All of the above

(h) For the following method of load flow analysis number of iterations does not depend on the size of the system under study
(i) Gauss Seidal method
(ii) Both of these
(iii) Newton Raphson Method
(iv) None of these

(i) The most accurate solution can be achieved by
(i) Gauss-Seidal method
(ii) Fast Decoupled method
(iii) Newton Raphson Method
(iv) Both (i) and (iii)

(j) In an admittance matrix if a mutual admittance element is zero it means
(i) There is no connection between those two buses
(ii) The line connecting those buses is purely resistive
(iii) There is a transformer connected between those buses
(iv) None of the above

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Q.2 (a) In a 3-phase, 4-wire system, the currents in R, Y and B lines under a fault condition are given as: $I_R = 100 \angle 30°$ A; $I_Y = 50 \angle 300°$ A; $I_B = 30 \angle 180°$ A. Calculate the positive, negative and zero sequence currents in the R-line and return current in the neutral wire? [8]

(b) Prove that in case of a delta connection, the zero sequence component of current has no effect on the line current. [6]

Q.3 (a) Derive Swing equation. [7]

(b) A 2pole, 50 Hz, 11kv turbo alternator has a rating of 100 MW, power factor 0.85 lagging. The rotor has a moment of inertia of 10,000 kgm^2. Calculate H and M. [7]

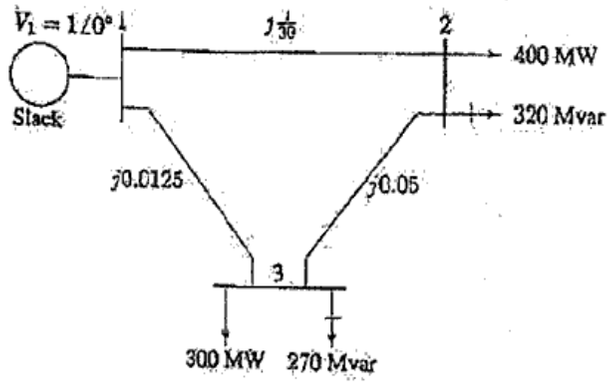
Q.4 (a) Write down the algorithm (steps) of Newton-Raphson power flow method. [8]

(b) What do you understand by contingency analysis? Explain the different states of the power system [6]

Q.5 (a) Derive the expression of voltage, active and reactive power used in load flow study by Gauss-Seidal method. [10]

(b) What is acceleration factor used in Gauss-Seidal method of power flow? What is its typical value and why it is used? [4]

Q.6 (a) Calculate the bus admittance matrix for the given system. The impedance values of the branches and the bus numbers are given in the diagram. [7]



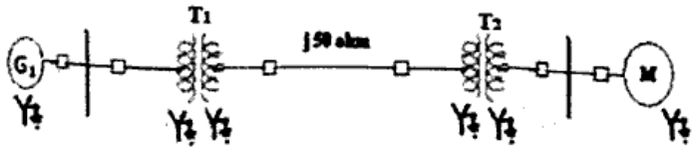
(b) For the 3-bus system given in Q6(a) what will be size of the Jacobian matrix? Write the expressions of the elements of the jacobian matrix. (Only expression in terms of V and Y elements are required). [7]

Q.7 (a) Describe the working principle of Static Var Compensator (SVC) with neat schematic diagram. [7]

(b) Explain the role of SVC on regulating AC voltage. [7]

Q.8 (a) What are the advantages of pu system representation in power system? How to find out the pu value of any given quantity? [6]

(b) Draw the reactance diagram for the power system shown in figure below. Use a base of 100 MVA, 220KV in the transmission line. The ratings of the generator motor and transformer are: Generator: 40MVA, 25KV, X=20% Motor: 50MVA, 11KV, X=30% Transformer T1: 40MVA, 33/220 KV, X=15% Transformer T2: 30MVA, 11/220 KV, X=15%. [8]



Q.9 (a) Explain the equal area criterion for a single machine connected to infinite bus.

(b) Write short note on – Automatic Generation Control (AGC) [7]

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